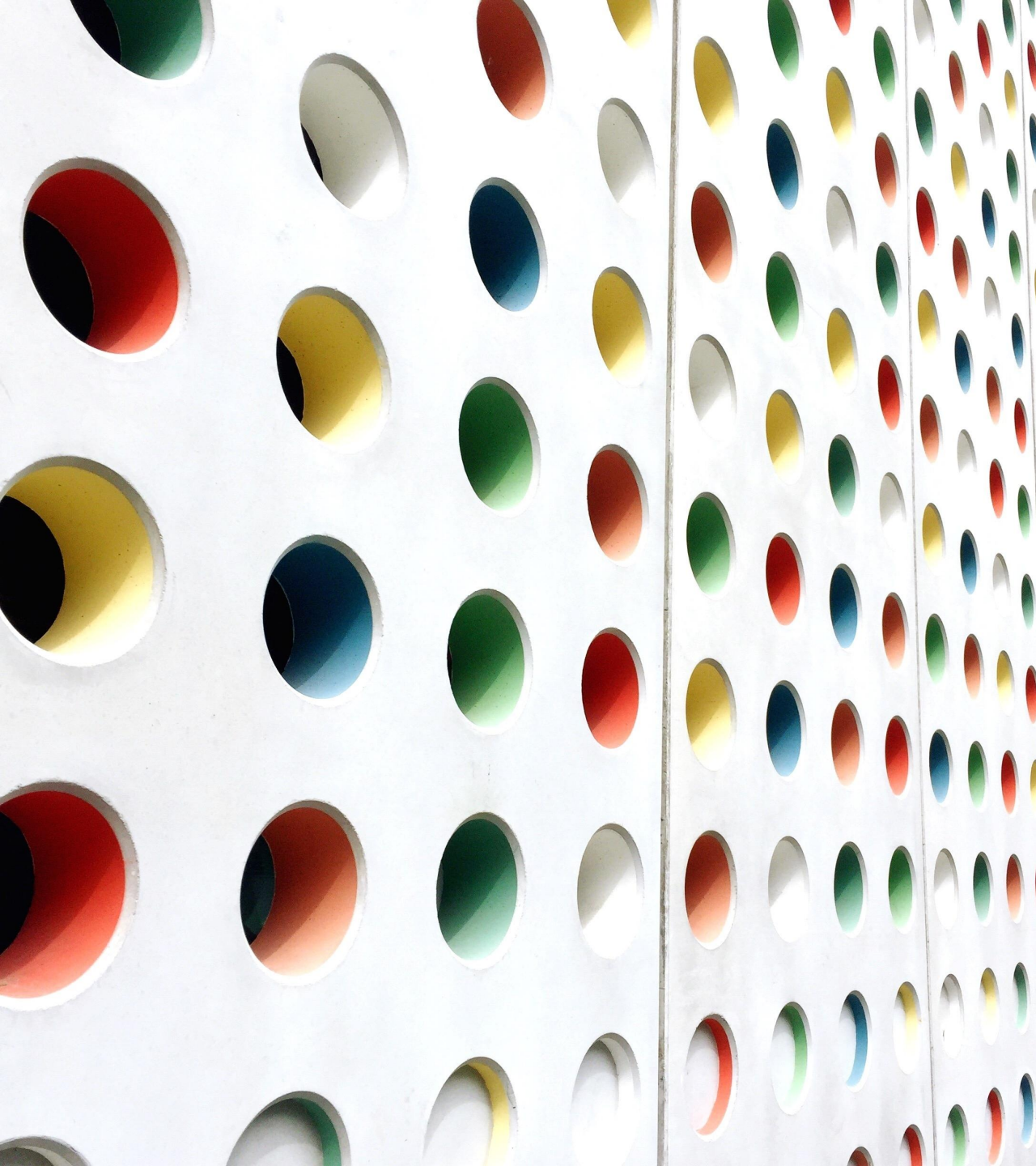




Semantic Annotation and Retrieval: RDF

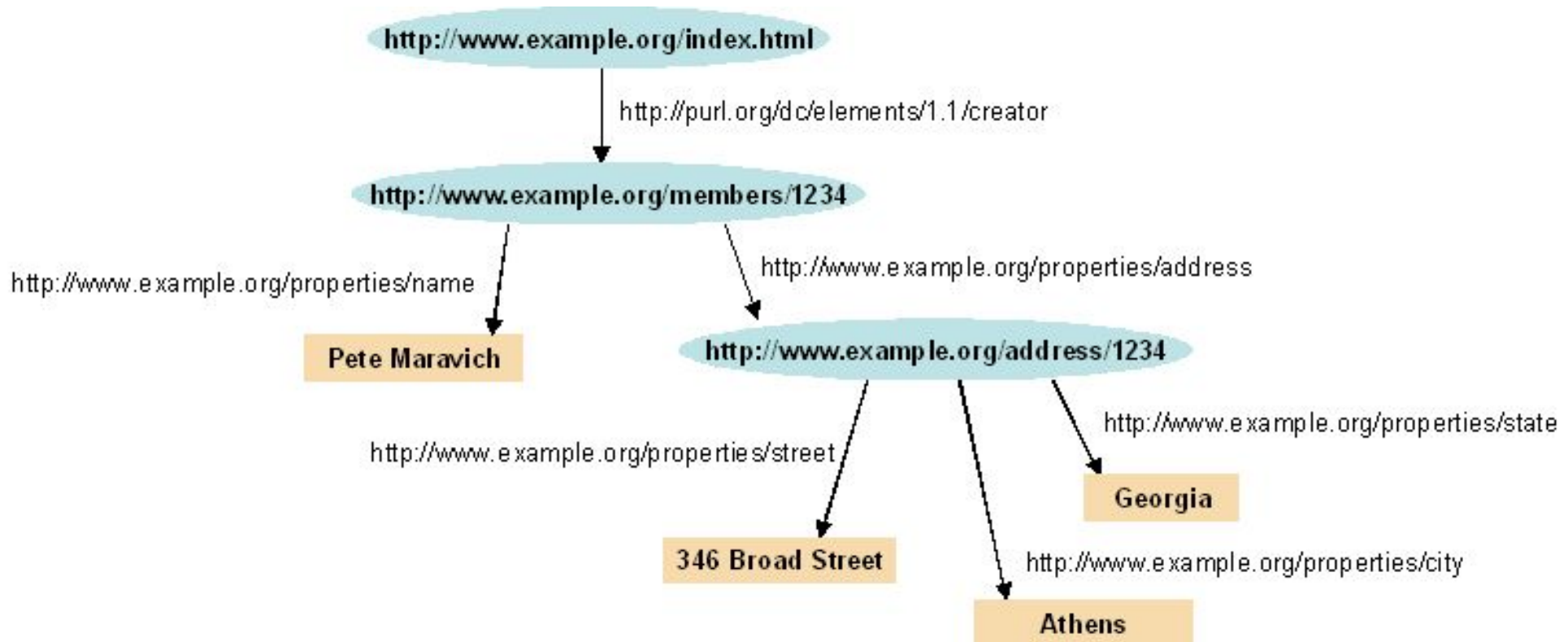
BY

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What is RDF?

- Resource Description Framework (RDF) is a framework to publish statements on the Web about anything. It allows anyone to describe resources, in particular Web resources, such as the author, creation date, subject, and copyright of an image.
- Any information portal or data-based website can be interested in using the graph model of RDF to open its silos of data about persons, documents, events, products, services, places, etc.
- RDF reuses the Web approach to identify resources (URI) and to allow one to explicitly represent any relationship between two resources. Such statements can come from any source on the Web and be merged with other statements supporting worldwide data integration.
- By using and reusing URIs, anyone can say anything about any topic, anyone can add to it, and so on.

URI



- *Resources* are a core concept on the Semantic Web: everything one might refer to is considered a resource; for example, a Web page, an image, videos, but also a person, a place, a device, an event, an organization, a product, or a service. More technically speaking, everything that can be identified by a URI can be considered a resource.
- *Descriptions* of resources are essential for understanding and reasoning about them. In the most general case, a description is a set of attributes, features, and relations concerning the resource.
- The *Framework* means it provides models, languages, and syntaxes for these descriptions.

Model of RDF

Principles for RDF

Define a new framework for viewing, manipulating, and associating networked collections of distributed information. As such, it provides interoperability between applications that exchange machine-understandable information.

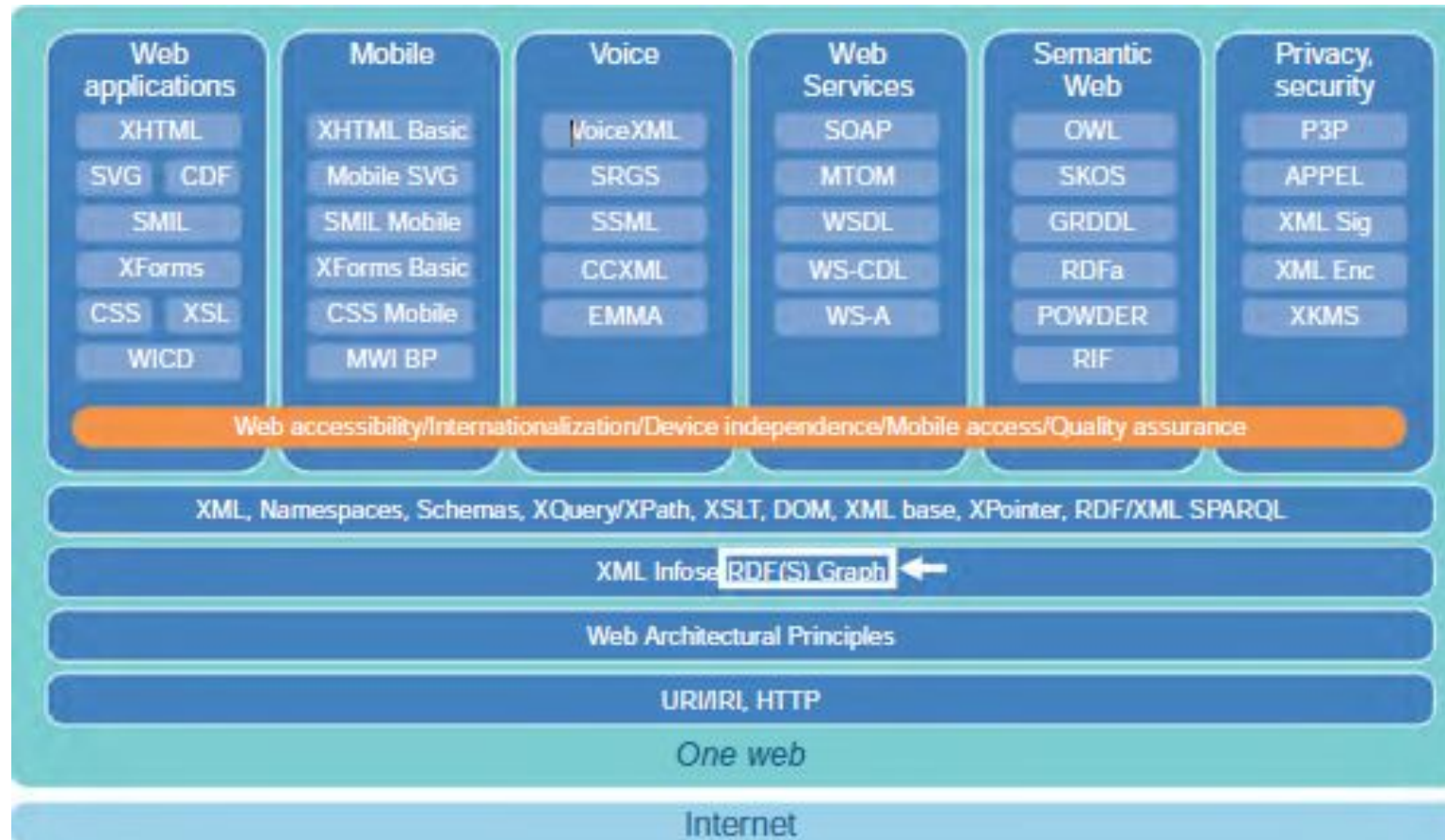
- RDF provides a standard data structure and a model to encode data and metadata about any subject on the Web.
- To this end;
 - Annotated resources on the Web are then connected to other annotated artifacts
 - Spanning a network of connected resources which links typed information
 - Discovering such implicit knowledge and making it explicitly available to interested parties via formal representations
 - Web applications is the main driver of current Semantic Web research
 - Besides a cloud of facts that are available through the Semantic Web, more and more ontologies are published that allow making sense out of the Web of Data

Key features of RDF

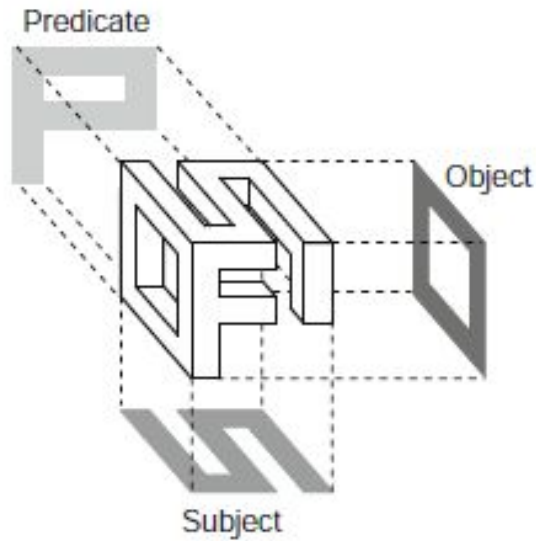
RDF not only provides a graph model to publish and link data on the Web, it also provides the fundamental shared data model on which other capabilities are built:

1. Querying (SPARQL is built on top of RDF)
2. Embedding (RDFa and GRDDL rely on the RDF model)
3. Reasoning (RDFS and OWL are defined on top of RDF)

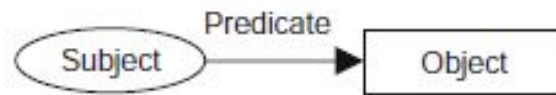
RDF graphs at the foundations of the recommendation stack at W3C



1. RDF Triple— an atom knowledge



(Subject, Predicate, Object)



- RDF graphs are constructed by RDF triples.
- RDF triples describe and connect objects via the combination of resources, properties, and property values; such triples are also referred to as statements.
- The resource is the subject of the statement, the property is the predicate of the statement, and the property value is the object of the statement.
- The basic data structure of RDF is a triple of the form <subject, predicate, object>.
- Examples: (doc.html, author, Fabien)
- Examples: (doc.html, theme, music)

2. Graph-Oriented Data Model

- RDF is a decentralized data representation model relying on distributed triples that form a global graph.
- The identity of the resources is based on the URI mechanism: if two resources have the same URI they are one and the same node in the graph.

RDF Graph

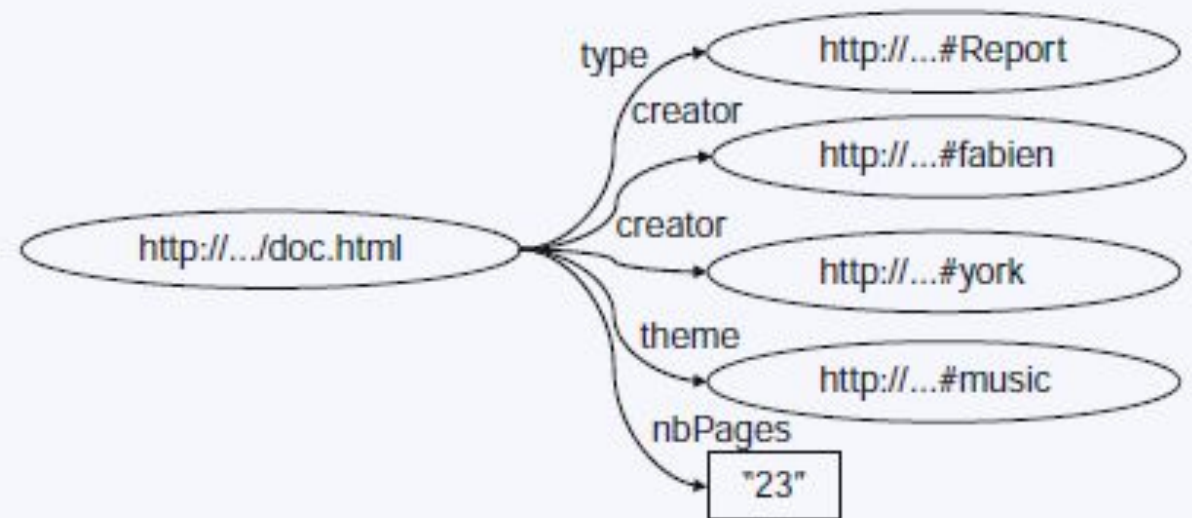
- A multi-graph: a graph that can contain both multiple edges and loops between its vertices
- A directed graph: every edge is oriented going from the end-vertex representing the subject to the end-vertex representing the object
- A labeled graph: edges are labeled with URIs, vertices are labeled with URIs, blank node identifiers, or literals

Simple RDF Example

English: "The report doc.html has the authors Fabien and York, the theme music and is 23-pages long."

Logic predicates:
Report(doc.html)
creator(doc.html,Fabien)
creator(doc.html,York)
theme(doc.html,Music)
nbPages(doc.html, 23)

Triples:
(doc.html, type, Report)
(doc.html, creator, Fabien)
(doc.html, creator, York)
(doc.html, theme, Music)
(doc.html, nbPages, "23")



3. Identifying Conceptual Vocabularies Through Namespaces

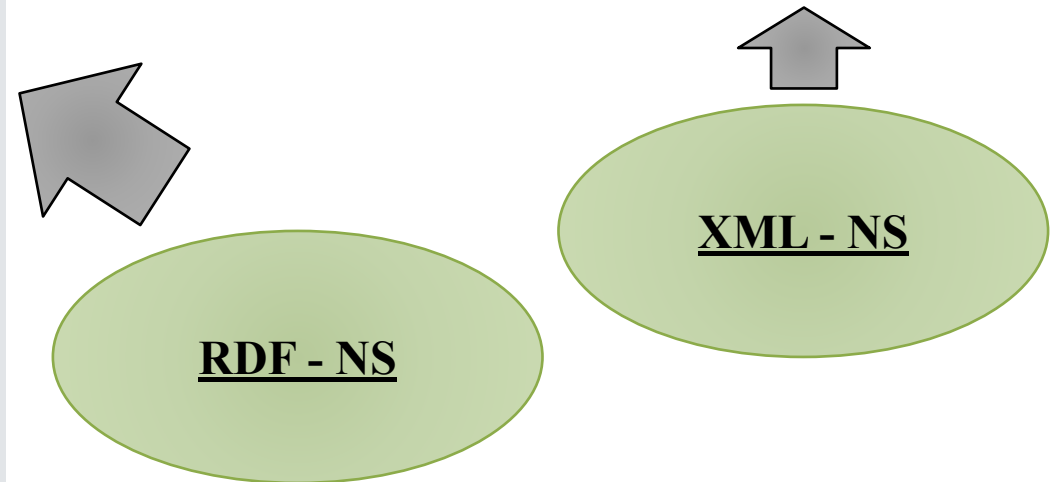
- When a URI is used to identify a set of terms, a vocabulary, a schema; in that case the URI is called a namespace.
- Namespaces are used to identify the schemas declaring the types of resources and types of relations used to label RDF graphs.
- For instance, RDF provides an elementary typing primitive type that belongs to the core RDF vocabulary which is identified by the URI

<http://www.w3.org/1999/02/22-rdf-syntax-ns#>.

- Consequently, the type predicate can be identified as `rdf:type` instead of

<http://www.w3.org/1999/02/22-rdf-syntax-ns#type>.

- In XML documents namespaces are associated to prefixes in order to shorten the resource identifiers in the document, locally using the prefix instead of the full URI.



● rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#
● rdfs	http://www.w3.org/2000/01/rdf-schema#
● foaf	http://xmlns.com/foaf/0.1/
● dc	http://purl.org/dc/elements/1.1/
● xsd	http://www.w3.org/2001/XMLSchema#
● exs	http://example.org/schema#
● exr	http://example.org#

All resources are identified with a URI and denoted by the short form **prefix : localname** of the namespace mechanism that is inherited from XML.

Namespaces and Associated Prefixes

Triples

```
(http://example.org/doc.html , rdf:type , exs:Report)  
(http://example.org/doc.html , dc:creator , exr:Fabien)  
(http://example.org/doc.html , dc:creator , exr:York)  
(http://example.org/doc.html , exs:theme , exr:Music)  
(http://example.org/doc.html , exs:nbPages , "23")
```

Rewritten example using namespaces and
URIs

4. Typing and Multi-instantiation

- RDF typing primitive `rdf:type` allows for attaching schema information in the form of one or more types to a resource: `http://example.org/doc.html` is declared as being a `report`: that is, a resource of type `exs:Report`.
- RDF allows for multi-instantiation -- RDF resource can belong to several unrelated classes that is the main difference between RDF and OOP.
- Primitive data values are mostly typed by means of the XML Schema datatypes.

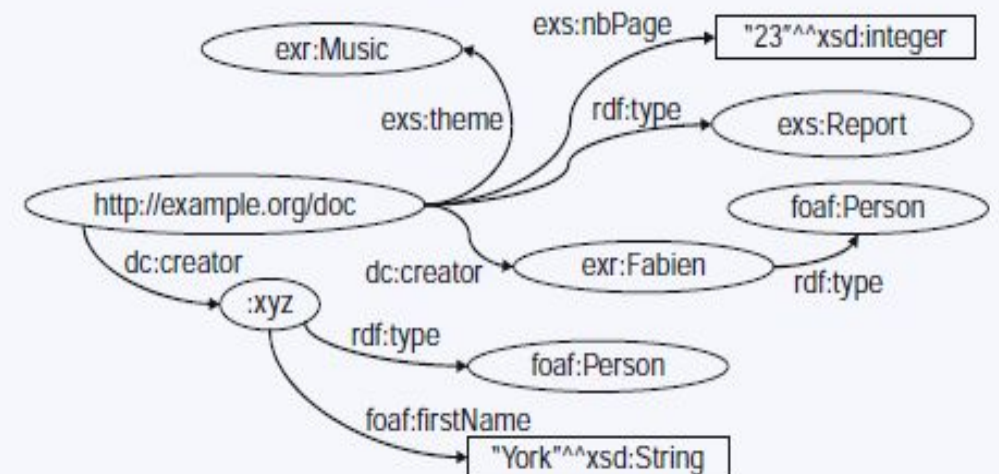
5. Existential Quantification of Resources: Blank Nodes

- In some cases this unified identification of resources is not possible. There are situations when only the type of object, not however a particular instance, is known, or when the instance is anonymous. The RDF response to this is so-called blank nodes.
- Blank node is given as **_:xyz**. The underscore represents the unknown namespace, and hence, the anonymous identifier does not assume a globally unique name.

Introducing typed literals and RDF blank nodes

Triples

```
(http://example.org/doc.html , rdf:type , exs:Report)
(http://example.org/doc.html , dc:creator , exr:Fabien)
(http://example.org/doc.html , dc:creator , _:xyz)
(_:xyz, foaf:firstName, "York"^^xsd:String)
(http://example.org/doc.html , exs:theme , exr:Music)
(http://example.org/doc.html , exs:nbPages , "23"^^xsd:integer)
```



Serializing RDF Graphs

- RDF graph notation provides an abstract model only that is not machine-interpretable.
- Being understandable by machines also implies being interoperable and exchangeable across distributed and heterogeneous systems and applications, and obviously the Web.
- Converting the abstract RDF graph into a concrete exchange format – W3C recommends standardized serialization mechanisms.
- Ways:
 - RDF Graphs as XML Trees
 - Triple Serialization in N-Triples
 - Triple Serialization in Turtle

1. RDF Graphs as XML Trees

- `<rdf:RDF>` – root element
- Attributes to the root element are used to specify namespace information and prefixes
- `<rdf:Description>` -- element with the attribute `rdf:ID` (with a simple ID that would be expanded to a full URI by means of the base namespace of the XML document) or `rdf:about` (with a URI) to identify resources and defines child elements (properties) to describe the resource.
- The references via `rdf:ID` or `rdf:about` allow the specification of graphs that are hidden in XML trees.
- `<rdf:Description>` and `<rdf:type>` elements could be merged as a class element for simplicity (e.g., `<exs:Report>`).
- Property elements may be declared as attributes of the description or as child elements. For-example: `exs:nbPages` property

RDF/XML serializations

RDF/XML serializations

```
<?xml version="1.0"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:exs="http://example.org/schema#">
  <rdf:Description rdf:about="http://example.org/doc.html">
    <rdf:type rdf:resource="http://example.org/schema#Report"/>
    <exs:theme rdf:resource="http://example.org#Music"/>
    <exs:theme rdf:resource="http://example.org#History"/>
    <exs:nbPages rdf:datatype="http://www.w3.org/2001/XMLSchema#int">23</exs:nbPages>
  </rdf:Description>
</rdf:RDF>
```

```
<?xml version="1.0"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:exs="http://example.org/schema#">
  <exs:Report rdf:about="http://example.org/doc.html" exs:nbPages="23">
    <exs:theme rdf:resource="http://example.org#Music"/>
    <exs:theme rdf:resource="http://example.org#History"/>
  </exs:Report>
</rdf:RDF>
```

2. Triple Serialization in N-Triples

- N-Triples provides a line-based, plain text format for serializing RDF graphs. N-Triples is designed to be a fixed subset of N3 and subset of Turtle.
- (Notation-3 (N3)), is an RDF serialization that also includes constructs such as variables and quantification for the integration of rules about the RDF facts.
- N3 allows for the quoting of graphs by means of { and }, in order to make statements about statements.

N-Triples yields a simplified serialization based on Turtle and N3.

Every triple is provided as subject, predicate, and object that are separated by whitespaces and terminated by a dot (.).

N-Triples serialization of the RDF graph in RDF/XML.

- Namespace `http://example.org/schema#` repeatedly appears in several resources and the same subject resource is also repeated in each statement.

```
<http://example.org/doc.html> <http://www.w3.org/1999/02/22-rdf-syntax-ns#type>  
  <http://example.org/schema#Report>.  
<http://example.org/doc.html> <http://example.org/schema#theme>  
  <http://example.org#Music>.  
<http://example.org/doc.html> <http://example.org/schema#theme>  
  <http://example.org#History>.  
<http://example.org/doc.html> <http://example.org/schema#nbPages>  
  "23"^^<http://www.w3.org/2001/XMLSchema#integer>.
```


RDFS

Define domain-specific classes and properties to describe these resources and organize them in hierarchies. These schemas are also published and exchanged in RDF.

RDF Schema (Resource Description Framework Schema, variously abbreviated as **RDFS**, **RDF(S)**, **RDF-S**, or **RDF/S**) is a set of classes with certain properties using the RDF extensible knowledge representation data model, providing basic elements for the description of ontologies, otherwise called RDF vocabularies, intended to structure RDF resources. These resources can be saved in a triplestore to reach them with the query language SPARQL.

RDFS Constructs – RDFS Classes

RDFS constructs are the RDFS classes, associated properties and utility properties built on the limited vocabulary of RDF.

Classes

- **rdfs:Resource** is the class of everything. All things described by RDF are resources.
- **rdfs:Class** declares a resource as a class for other resources.

An example of an **rdfs:Class** is **foaf:Person** in the Friend of a Friend (FOAF) vocabulary. An instance of **foaf:Person** is a resource that is linked to the class **foaf:Person** using the **rdf:type** property

Following formal expression of the natural-language sentence : 'John is a Person'.

ex:John rdf:type foaf:Person

The definition of **rdfs:Class** is recursive: **rdfs:Class** is the class of classes, and so it is an instance of itself.

rdfs:Class rdf:type rdfs:Class

The other classes described by the RDF and RDFS specifications are:

- **rdfs:Literal** – literal values such as strings and integers. Property values such as textual strings are examples of RDF literals. Literals may be plain or typed.
- **rdfs:Datatype** – the class of datatypes. **rdfs:Datatype** is both an instance of and a subclass of **rdfs:Class**. Each instance of **rdfs:Datatype** is a subclass of **rdfs:Literal**.
- **rdf:XMLLiteral** – the class of XML literal values. **rdf:XMLLiteral** is an instance of **rdfs:Datatype** (and thus a subclass of **rdfs:Literal**).
- **rdf:Property** – the class of properties.

Properties are instances of the class `rdf:Property` and describe a relation between subject resources and object resources. When used as such a property is a predicate.

- **rdfs:domain** of an `rdf:Property` declares the class of the *subject* in a triple whose predicate is that property.
- **rdfs:range** of an `rdf:Property` declares the class or datatype of the *object* in a triple whose predicate is that property.

For example, the following declarations are used to express that the property `ex:employer` relates a subject, which is of type `foaf:Person`, to an object, which is of type `foaf:Organization`:

```
ex:employer    rdfs:domain    foaf:Person
ex:employer    rdfs:range     foaf:Organization
```

Given the previous two declarations, from the triple:

```
ex:John        ex:employer    ex:CompanyX
```

can be inferred (resp. follows) that `ex:John` is a `foaf:Person`, and `ex:CompanyX` is a `foaf:Organization`.

- **rdf:type** is a property used to state that a resource is an instance of a class. A commonly accepted qname for this property is "a".
- **rdfs:subClassOf** allows declaration of hierarchies of classes.

For example, the following declares that 'Every Person is an Agent':

```
foaf:Person    rdfs:subClassOf    foaf:Agent
```

Hierarchies of classes support inheritance of a property domain and range (see definitions in next section) from a class to its subclasses.

- **rdfs:subPropertyOf** is an instance of `rdf:Property` that is used to state that all resources related by one property are also related by another.
- **rdfs:label** is an instance of `rdf:Property` that may be used to provide a human-readable version of a resource's name.
- **rdfs:comment** is an instance of `rdf:Property` that may be used to provide a human-readable description of a resource.

RDFS

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